Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/05/2008
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Project Title:	Impact of HZE Particles Exposure on Proliferation and Differentiation of Human Neural System Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per J. Dardano/JSC (5/09) NOTE: Received NCE to 5/31/2009 per J. Dardano/JSC (5/08) End date changed back to 6/1/2008 per PI (5/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chuang, Eric (National Institutes of Health) Morrill, Steve (UTMB) Wu, Ping (UTMB)		
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Task Description:	Space radiations such as HZE particles pose a great risk of damaging the central nerve system (CNS) of astronauts. Human neural stem cells (hNSCs) are of particular interest since functional NSCs in adult brain are potential targets of HZE damage, and because exogenous hNSCs may provide potential therapeutic source for replacing lost neurons in radiation-damaged brain. The long-term objective of this project is to study the mechanisms underlying HZE particles-induced adverse impact on the proliferation and differentiation of a primary human fetal neural stem cell line, K048. K048 retains self-renewal and multipotential differentiation capabilities in vitro and in vivo. Moreover, a novel in vitro priming procedure appears to direct K048 cells to adopt neuronal differentiation fate. We will test two hypotheses:(1) HZE particles disrupt proliferation and differentiation of human NSCs by activating apoptotic and genotoxic-response pathways. (2) Inhibitory neurogenic micro-environmental cues produced by HZE in rat brain can be overcome by this novel in vitro priming prior to transplantation into brain. We will pursue three specific aims: (1) Determine the impact of 56Fe and gamma-ray irradiation on proliferation and differentiation of K048 cells in vitro. (2) Determine gene expression patterns and apoptotic responses of proliferating and differentiating K048 cells following 56Fe and gamma-ray irradiation. (3) Determine whether in vitro primed K048 cells can differentiate into neurons when grafted into neurogenic and non-neurogenic regions of rat brains damaged by 56Fe and gamma-rays. The results of this project will contribute to a better understanding of cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles. To determine the impact of HZE on proliferation and differentiation of K048 neural stem cells in vitro. To determine the fate of primed K048 cells grafted into 56Fe-irradiated rat brains. In this project, we propose to examine the impact of HZE and low LET radiations on the proliferation and differentiation of K048 human neural stem cell line. The K048 cell model is a unique experimental system not only because of its human origin, but also because this line is not transformed and retains the characteristics of normal neural stem cells. Using this system we will address directly to one of the high priority topics listed in NRA 03-OBPR-07 Research Announcement, namely “What are the cellular and molecular mechanisms of damage to the CNS following irradiation by HZE particles?”. The proposed research is also within the scope of “Critical Path Roadmap”, regarding risk number (39) “Degenerative tissue diseases including damage to the central nervous system, cataracts, circulatory diseases, or other late non-cancer risks”, and risk number (41) “Early or acute effects from radiation exposure”. Specifically, our proposed studies address Critical Question (10.05) in radiation, “Are there unique biological effects associated with HZEs?”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Space radiations such as HZE particles pose a great risk of damaging the central nerve system (CNS) of astronauts. Human neural stem cells (hNSCs) are of particular interest since functional NSCs in adult brain are potential targets of HZE damage. In this NASA funded project, we use a human fetal neural stem cell line K048 to test two hypotheses: (1) HZE particles disrupt proliferation and differentiation of human NSCs by activating apoptotic and genotoxic-response pathways. (2) Inhibitory neurogenic micro-environmental cues produced by HZE in rat brain can be overcome by this novel in vitro priming prior to transplantation into brain. The K048 cells retain self-renewal and multipotential differentiation capabilities in vitro and in vivo, and thus provide a unique experimental model system not only because of its human origin, but also because this line is not transformed and retains the characteristics of normal neural stem cells. In the previous reporting periods, we have studied the does-response and timing-response relations in 56Fe ion-induced inhibition of neuronal differentiation of K048 cells. We have found that 1 GeV/n heavy iron ions inhibit neuron generation from stem cells in a dose-dependent manner, and this effect is persistent several passages post irradiation. On the other hand, 56Fe irradiation appears to favor astroglial differentiation, but this effect diminishes gradually post irradiation. In this reporting period, we have studied how 56Fe-irradited rat brain affect differentiation of grafted human neural stem cells. We irradiated SD rat brain with 3 Gy of 1 GeV/n 56Fe ions. Eight to ten weeks post-irradiation, a marked microgliosis in the brain was evident, indicating a neuroinflammation status. At this point, we labeled K048 hNSCs with green fluorescent protein (GFP), and then transplanted these cells into SD rat brains. One month after grafting, animal were sacrificed and the fate of transplanted hNSCs were examined by immunohistochemical staining. We found that 56Fe-irradiated rat brains exhibit an inhibitory effect on neuronal differentiation of grafted cells. However, an in vitro priming treatment of these K048 cells with a combination of growth factors prior to transplantation appears to enhance the neuronal differentiation of K048 cells in rat brain in general, and can counteract the inhibitory effect of neural inflammation in particular. Our results support the notion that the effects of space radiation on neural stem cells and underlying mechanisms need to be examined in the context of interactions between stem cells and glial cells. Thus, we will continue our current study by examine the effect of irradiated microglial and astroglial cells on neuronal differentiation of hNSC in vitro.
Bibliography Type:	Description: (Last Updated: 03/19/2010)
Abstracts for Journals and Proceedings	Yu Y, Gao Y, Wu P. "Impact of Fe56 ion radiation on human neural stem cell differentiation." 18th Annual NASA Space Radiation Investigators’ Workshop, Rohnert Park, California, July 13-15, 2007. Abstracts, 18th Annual NASA Space Radiation Investigators’ Workshop, Rohnert Park, California, July 13-15, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Yu Y, Gao J, Gao Y, Dunn T, Wu P. "Neuronal differentiation of grafted human neural stem cells in 56Fe-irradiated rat brain." NASA Human Research Program Investigators' Workshop, League City, Texas, February 4-6, 2008. Abstracts, NASA Human Research Program Investigators' Workshop, League City, Texas, February 4-6, 2008. , Feb-2008